Valmik Nahata

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I'm Valmik Nahata, though most people know me as Val. I am an undergraduate at the University of California, San Diego studying Data Science and Cognitive Science. I conduct research in medical AI at Massachusetts General Hospital and was previously at Dartmouth Health. I work across (1) Medical AI, (2) AI Safety & Governance, and (3) AI Evaluations & Scientific Computing, with the overarching goal of building foundation models and evaluation frameworks that generalize and democratize science for advancing humanity.

Work

Massachusetts General Hospital (Harvard affiliate) *Nov 2025 – Present*
Undergraduate Researcher

Dartmouth Health *Jun 2024 – Sept 2024*
Research Assistant

Education

University of California, San Diego *Sept 2024 – Present*
B.S. in Data Science & Cognitive Science
Provost Honors

Papers & Research in Progress

Google Scholar: I'm early in my research career, with a few works on Google Scholar and more in preparation.
H-Index: 2
Citations: 9

Medical AI

[1] Self-reflection and answer revision in clinical language models (2026). *Research in progress.* I study what happens when a language model is asked to review its own diagnosis, testing whether reflection prompts surface errors or instead elicit critiques the model then acts on, whether or not they are valid. By branching every intervention from the same initial diagnosis on real clinical cases, I isolate the effects of asserting an error, requesting a critique, varying a critique's author and relevance, iterating review, and supplying withheld clinical evidence, with implications for the critic and verifier agents now built into clinical AI systems.

[2] MedARC Agentic Medical Fact Verifier (2026). *Research in progress.* I developed scrapers for MedlinePlus, StatPearls, and USPSTF to build the medical evidence corpus. I also contributed to evaluation of claim decomposition, evidence retrieval, and fact verification.

[3] Inference Time Adaptation of a Surgical Video Language Model (2026). *Accepted at the ECCV MedVidU Workshop.* I studied inference-time adaptation of a frozen Qwen2.5-VL-based surgical model across eight MedVidBench tasks, focusing on temporal grounding and surgical captioning through explicit video timing, structured output requirements, and in-context demonstrations.

– Medical Video Challenge Runner-up, ECCV MedVidU Workshop 2026

[4] Shivashankar, A., Nair, T., **Nahata, V.**, Friedman, C., Azher, Z., & Levy, J. J. (2024). Retrieval Augmented Generation for Pathology Reports. Dartmouth Health EDIT AI Conference poster.

AI Safety & Governance

[1] MAESTRO4IHL: A Threat Model Framework for Synthetic Media Harms to POW Protections under IHL (2026). *Accepted at the NeurIPS AI for Peace Workshop.* We developed a seven-layer threat model mapping synthetic media attacks on prisoner-of-war records, communications, and legal protections to technical mitigations and governance responsibilities. The framework was previously presented by my coauthors at UNIDIR's recent Global Conference on AI, Security and Ethics in Geneva.

[2] Simple Features Beat Most AI Control Monitors (2026). *Accepted at the NeurIPS EvoRobust Workshop.* We studied authorship and problem-selection confounds in AI control benchmarks using static code and problem features. We regenerated honest and backdoored programs with the same model and matched prompts, checked backdoors against public tests, input constraints, and reference solutions, and re-evaluated monitors after removing invalid examples.

[3] Dorn, F., Ferrés, X., Jang, E., & **Nahata, V.** (2025). The Early Economic Impacts of Transformative AI: A Focus on Temporal Coherence. Economics of Transformative AI Sprint report, Apart Research & BlueDot Impact.

– First Prize, Apart Research & BlueDot Impact Economics of Transformative AI Sprint 2025

AI Evaluations & Scientific Computing

[1] Massive Omni-Embedding Benchmark & MTEB (2026). *Research in progress.* I contributed 18 MTEB evaluation tasks across eight datasets for cross-modal retrieval across text, speech, images, and video, emotion classification, and acoustic-scene clustering, implementing dataset construction, modality isolation, and speaker-disjoint splits, and evaluated CLAP, X-CLIP, CLIP, and Jina embedding models.

[2] PsychLing 101 (2026). *Research in progress.* I integrated the OneStop eye-movement dataset, the Form Priming Project, and the Provo Corpus to compare language models with human reading and priming behavior.

[3] HydroTuring (2026). *Research in progress.* I developed five model proposals for evaluating physical consistency in hydrological AI through comparisons of learned and physics-based models and conservation tests.

[4] Last Translation Benchmark (2026). *Preprint.* I contributed evaluation examples and verification rules for the benchmark's v2 update.

[5] Bader, D. A., Li, F., Du, Z., Pauliuchenka, P., Alvarado Rodriguez, O., Gupta, A., Minnal, S. S. V., **Nahata, V.**, Ganeshan, A., Gundogdu, A. C., & Lew, J. (2025). Cover Edge-Based Novel Triangle Counting. *Algorithms*, 18(11), 685.

[6] Doskoch, G. M., Basuroski, A., Halley, K., Sookram, A., Rodriguez-Ramos, I., **Nahata, V.**, et al. (2024). A Statistical Analysis of Crab Pulsar Giant Pulse Rates. *The Astrophysical Journal*, 973(2), 87.

Awards

Poseidon Fellowship — *UC San Diego, Apr 2026*

Machine Learning Challenge Winner — *UC San Diego & NSF HDR, Jan 2026*

Hackathon Finalist — *Milwaukee Bucks, Mar 2025*

Press & Newsletter Mentions

- **Syntony:** Two research presentations accepted for NeurIPS 2026 in Paris (2026).
- **UC San Diego:** Poseidon Fellows celebrates its first cohort of student innovators (2026).

Service & Field Building

- I am a member of MIT AI Alignment and was involved in the Columbia AI Alignment Club Summer Paper Reading Club.
- I previously served as Vice President of Research at the AI Collective at UC San Diego.
- I founded Labry, a nonprofit that connected students with research labs and opportunities to enter research.